FED. STD. 00152(Army-Ord)
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FEDERAL STANDARD

STEEL BAR, HOT ROLLED (INCLUDING BAR SIZE SHAPES)

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FOREWARD

The intent of this standard is to provide the best available guidance in the selection, procurement and use of commercial steel products. This standard is one of a series pertaining to steel mill products developed to accomplish this objective.

Hot rolled steel bars selected and specified in accordance with this standard will promote and ensure several important advantages:

(a) More experience in and application of commercial qualities and grades of steel mill products.

(b) Considerable reduction in the use of special steel requirements.

(c) Conservation of alloying elements and steel mill facilities.

(d) Reduced lead time.

(e) Procurement of the required quality of steel mill products at the lowest cost to the Government.

It is recognized that certain applications will demand additional or more stringent requirements than presented in this standard. Individual specifications will of necessity have to be developed, but in the development, this standard should be consulted. The basic qualities of this standard may be used as the starting point for any individual specification. Therefore, by referencing the applicable designations in this standard and adding the requirements that are not covered, but needed, a simple, supplemental specification will result.

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FEDERAL STANDARD

STEEL BAR, HOT ROLLED (INCLUDING BAR SIZE SHAPES)

1. SCOPE

1.1 Scope.- This standard covers industry practices and descriptions relating to hot rolled steel bars, as rolled or thermally treated, and as rolled bar size shapes in the following kinds of steels commonly produced as recognized qualities:

PART A - CARBON	PART C - HIGH STRENGTH LOW ALLOY
PART B - ALLOY	PART D - STAINLESS & HEAT RESISTING

1.2 Purpose.- The purpose of this standard is to provide engineering guidance to designers, materials engineers and contractors in the selection of hot rolled steel bars and bar size shapes consistent with commercial practices.

Steel bars are selected for use in terms of the kind of steel (alloying elements), the quality, condition and specified properties, restrictions, tests or other requirements. They are also generally priced accordingly. The kind of steel, the quality and other requirements specified for a particular application should therefore be limited to those necessary to meet the design requirements and fabrication methods.

The qualities described in each part of this standard are arranged generally in order of increasing higher quality and cost. Each quality is individually and separately identified. Also separately identified are such properties and requirements as composition, mechanical properties, grain size, condition (annealed, etc.), surface, hardenability, and decarburization limits. Because of this arrangement it is possible to specify a base quality steel bar containing a minimum of extras, e.g., "A-1" identifies a Merchant Quality carbon steel bar specified to composition in which a choice of several different compositions are provided; "A-5" identifies a Special Quality carbon steel bar specified to mechanical properties and providing a choice of eight strength levels. It is likewise possible to specify a base quality steel bar with one or more added

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compatible requirements, e.g., "A-5 and A-15" identifies a Special Quality carbon steel bar with special soundness requirement; "B-1, B-2, and B-11" identifies a Regular Quality alloy steel bar specified to composition, grain size and stress relieved condition.

Care should be exercised to avoid specifying a requirement not included in a specific quality description because this invariably and automatically raises the quality and the cost, e.g., to specify "check analysis" on a Merchant Quality carbon steel bar automatically invokes a Special Quality bar requirement, including a price increase.

Fabricating methods such as bending, forming, welding, punching, flanging, etc., can significantly change the specified properties or other conditions of the material, as received. For important or highly stressed parts, the effects of fabrication on properties of the finished part must be considered in the design, material selection and fabrication, or unsatisfactory performance may result.

It is not the intent of this standard to give design criteria or design allowables such as allowable design stresses. It is the intent to provide a means whereby standard commercial products can be selected and specified wherever they can be used to meet the design requirement. It is intended for reference on drawings and in product specifications; or it may be used directly in purchasing, if the kind of steel, the quality or identified application, and the composition, mechanical properties or other requirements, as applicable and described herein, are stated.

In the procurement of steel bars, the preparation and presentation of the ordering data by the procuring agency is of primary importance to ensure that the requirements for the steel bar are obtained at the lowest cost. The order of presentation in this standard will greatly facilitate the presentation of this data. In specifying steel bars the choice is first made of the quality desired, then the properties, chemical or mechanical and finally the additional requirements on production, properties, tolerances, etc., as required.

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PART A - STEEL BAR, CARBON, HOT ROLLED

1. SCOPE

1.1 Part A of this standard covers quality descriptions and requirements relating to hot rolled carbon steel bars as rolled or thermally treated, and as rolled bar size shapes. Mechanical property requirements, steel grades (compositions), machining allowances, sizes and size ranges, size tolerances, sampling for tests, inspection and definitions of terms are also included.

2. REFERENCED DOCUMENTS

2.1 The issues of the following documents in effect on the date of invitation for bids form a part of this standard to the extent specified herein.

FEDERAL STANDARDS

FED-STD-NO. 66 - STEEL: CHEMICAL COMPOSITION AND HARDENABILITY

FED. TEST METHOD STD. NO. 151 - Metals; Test Methods

NONGOVERNMENTAL STANDARDS

ASTM A273-58T - CARBON STEEL BLOOMS, BILLETS, AND SLABS FOR FORGING

3. DEFINITIONS

3.1 Bar Size Shapes.- Angles, channels, tees and zees are considered to be bars when their greatest sectional dimension is less than 3 inches. Sections 3 inches and larger are considered to be structural sections.

3.2 As Rolled.- As Rolled is a term which is used to denote bars containing the original mill scale and not given any type of thermal treatment. It is customary to furnish bars in the "as rolled" condition if no other processing is specified.

3.3 Ladle Analysis.- Ladle Analysis is the term applied to the chemical analysis representative of a heat or blow of steel and is the analysis reported to the purchaser. For Bessemer nonresulphurized steel, determinations for phosphorus and sulphur are made at intervals not greater than each 10 blows and the previous determination for phosphorus and sulphur is reported.

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3.4 Check Analysis.- Supplementary to ladle analysis, check analysis is a chemical analysis of steel in the semifinished or finished product form. It is not used, as the term might imply, for a duplicate determination made to confirm a previous result. The steel can be subjected to check analysis by the purchaser either for the purpose of verifying that the chemical composition is within specified limits for each element, including applicable tolerance for check analysis, or to determine variations in composition within a heat, blow or lot. The results of analyses representing different locations in the same piece or taken from different pieces of a heat, blow or lot may differ, within permissible limits, from each other and from the ladle analysis.

3.5 Thermal Treatment.- Heating and cooling steel in a way that will obtain desired conditions or properties. Heating and cooling for the sole purpose of mechanical working are excluded from this definition.

3.6 Ordinary Annealing.- Ordinary Annealing is the term generally applied to a thermal treatment the principal object of which is to soften the steel involved. Actual treatment cycles vary among producers, but in general the sheet is heated to a suitable temperature and slowly cooled. A compatible maximum hardness can be specified, subject to agreement with the producer.

3.7 Lamellar Pearlitic Annealing.- Lamellar Pearlitic Annealing involves heating to a suitable temperature followed by retarded cooling, to obtain a structure consisting of a lamellar aggregate of ferrite and iron carbide. The main purpose is usually to obtain improved machinability in certain grades.

3.8 Spheroidize Annealing.- Spheroidize annealing consists in a prolonged heating of steel at an appropriate temperature, followed by slow or cycle cooling to produce a globular or spheroidal condition of the carbide. This treatment produces a structure that is desirable in certain grades for machining or cold forming, for cold drawing operations, or for the effect it will have on subsequent heat treatment.

3.9 Stress Relief Annealing.- Stress relief annealing involves heating to a temperature below the critical range and holding for a time interval sufficient to equalize the temperature throughout the piece. The object of this treatment is to reduce stresses induced by machining, cold working or welding, or to restore elastic properties.

3.10 Normalizing.- Heating a ferrous material to a suitable temperature above the transformation range and then cooling in still air to room temperature.

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3.11 Quenching and Tempering.- In the more usually accepted terminology, heat treatment consists of three operations: (1) heating the steel above the critical range to approach a uniform solid solution (Austenite), (2) hardening by quenching in oil, water or air, and (3) tempering by reheating to a temperature below the critical range to secure the desired combination of strength and ductility.

3.12 Quenching.- Rapid cooling from a suitable austenitizing temperature, by contact with liquids or gases.

3.13 Tempering.- Reheating a quench hardened or normalized ferrous alloy to a temperature below the transformation range, and then cooling at any rate desired.

3.14 Hardening.- Increasing the hardness by suitable treatment, usually involving heating and cooling. When applicable, the following more specific terms should be used: Case Hardening, Flame Hardening, Induction Hardening, and Quenching.

3.15 Case Hardening.- Treating a ferrous part so that the outer portion or case is made substantially harder than the inner portion or core. Typical processes used for case hardening are Carburizing, Induction Hardening, and Flame Hardening.

3.16 Flame Hardening.- Flame Hardening consists of heating ferrous material, by means of a flame, to a depth and temperature which will result, depending on the cooling rate, in hardening of all or portions of the material.

3.17 Induction Hardening.- Induction Hardening consists of heating a ferrous material by means of electrical induction to a depth and temperature which will result, depending on the cooling rate, in hardening of all or portions of the material.

3.18 Carburizing.- Introducing carbon into a ferrous part by holding above the transformation range in contact with a suitable carbonaceous medium for a period of time required to develop the desired depth of case. The carburized part is usually quenched to produce a hardened case.

3.19 Decarburization.- The loss of carbon from a ferrous alloy as a result of heating in a medium that reacts with the carbon. The degree of decarburization decreases from surface to interior. At the surface there may be complete loss of carbon, leaving a zone of ferrite. The next zone may have only a partial loss with the carbon gradually increasing inwardly to the normal carbon content of the steel.

3.20 Subject to Agreement.- Subject to Agreement is used in numerous places in this standard. Agreements may be reached by developing a

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specific product specification in collaboration with industry or by making individual agreements with the producer at the time a contract is placed.

4. GENERAL REQUIREMENTS

4.1 Applications and Uses.- To provide for the economical use of hot rolled carbon steel bars they are specified in various qualities, restrictive requirements and qualities for identified uses. For minimum quality requirements Merchant Quality should be specified. Special Quality is intended for a number of applications as indicated below. For some more critical methods of fabrication and end use requirements Restrictive Requirements or Qualities for Identified applications may be specified (5.2.10). It is desirable to specify the end use when ordering steel bars, if possible.

4.2 Merchant Quality.- Bars of this quality are specified for a wide range of uses, such as structural and similar miscellaneous applications involving mild cold bending, mild hot forming, punching and welding, as used in the production of noncritical parts of bridges, buildings, ships, agricultural implements, road building equipment, railway equipment and general machinery.

4.3 Special Quality.- Special Quality is a fundamental quality, and can be specified with or without Restrictive Requirements, Restricted Requirements, or for Identified Applications. Applying each of these classes the consumer may specify chemical composition, mechanical properties, or thermal treatment, as defined in the body of this standard. Hot rolled carbon steel bars of this quality are intended for applications that may involve forging, heat treating, cold drawing, machining, and other usages.

4.4 Steel Making Process.- The steel making process shall be at the producer's option; however, the process used and practices followed shall be such as to provide reasonable assurance that the requirements of the specified Chemical Composition, Quality, other Requirements or Identified Application are met.

4.5 Chemical Composition.

4.5.1 Ladle Analysis.- The ladle analysis term (described in 3.3) applies to the limits and ranges of chemical composition, listed as standard, nonstandard, and restricted grades (5.2.9), which are used for specifying purposes. Standard steel grades (compositions) shall be identified by the AISI steel members. The ladle chemical ranges and limits for carbon steels shall be as shown in Federal Standard 66 and this standard.

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(1) Standard Nonresulphurized Carbon Steels
(2) Standard Resulphurized Carbon Steels
(3) Standard Rephosphorized and Resulphurized Carbon Steels
(4) Standard Acid Bessemer Resulphurized Carbon Steels
(5) (Standard) Ranges and Limits for Other Than Standard Carbon Steels
(6) Merchant Quality Chemical Compositions are described in Tables II, III, and IV.

4.5.2 Check Analysis.- Check analysis tolerances, for other than Merchant Quality, shall be in accordance with Fed. Std. 66 for Check Analysis Tolerances for Carbon Steel Bars. Merchant Quality bars and rimmed and capped steels are not subject to check analysis. If specified by the contract and requested by the Government, a set of drillings shall be provided by the producer for analysis by an identified laboratory.

4.5.3 Incidental Elements.- Incidental elements which occur in the normal melting of a heat may have an influence on the characteristics of the final product. Therefore, reporting of the analysis of specific elements may be specified. When necessary a limit prescribing the maximum allowable percentage of specific elements may be negotiated except for Merchant Quality.

5. DETAILED REQUIREMENTS

5.1 Merchant Quality.

5.1.1 Procurement Guide.- A guide to specifying requirements in Merchant Quality is presented in Table I.

This guide has been prepared to indicate those requirements which form a part of the basic quality description and in addition those additional requirements which must be specified by the procuring agency or negotiated with the producer. Additional requirements appended to or restricting the basic quality requirements are reflected in the cost of the product.

As an example, if Merchant Quality bar could be used for an application, a choice could be made between ordering to chemical composition or mechanical properties. If the former, one of the methods shown in the guide and denoted with X1 would be selected for ordering. By running down the listing in the guide all the requirements marked A would apply to bars

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ordered to this chemistry. If the application of these bars made it economic to use specific lengths rather than random lengths as would be furnished with the basic quality, this additional requirements would be specified.

This guide may also be used by the designer to determine if basic quality steels will satisfy the design parameters to avoid overspecification.

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TABLE I

GUIDE TO SPECIFYING REQUIREMENTS IN MERCHANT QUALITY HOT ROLLED CARBON STEEL BARS IN THE FORM OF ROUNDS, SQUARES, HEXAGONS - UNDER 3" FLATS AND OTHER SECTIONS UNDER 12 SQ. IN. (40.8 LBS./FT.)

	Requirement	Condition - As Hot Rolled	
		Chemical Composition	Mechanical Properties
Ladle Chemical Composition	Standard Steels	X1	
	Standard Steels C .50 Max., Mn .60 Max.	X1	
	"MQ" Steels	X1	
	Max. C .37/.50 (10 Pt. Min. Range) Mx .25/.60	X1	
	Max. C .35/.50 (8 Pt. Min. Range) Mn .30/.60	X1	
	Cu .20 Min.	S	S
	Two Elements - P, S		A
Mechanical Properties	Tensile Test, Standard ..(Table.I).....		X3
	Tensile and Cold Bend Tests .(Table.I.&.II)..		X3
	Tensile Strength, Min., Max., or Range .. To 95,000 psi Max.		X3
Type of Steel	Non-killed Steel	A	A
Size	Standard	A	A
	Nonstandard	N	N
Cutting Length	Standard Mill Cutting	A	A
	Special Re-cutting of Bar Ends	S	S
	Random Lengths	A	A
	Specified Length - Max. Shorts 10%	A	A
	Multiple Lengths	N	N
	Specified Length - No Shorts	S	S
Tolerances	Straightness Tolerances, Standard	A	A
	Straightness Tolerances, Special	S	S
	Dimensional Tolerances, Standard	A	A
	Length Tolerances, Standard	A	A
Processing	Pickled, Coated or Uncoated	S	S
Report of Tests	Ladle Analysis	A	A
	Tensile Test	-	A
	Bend Test	-	A

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TABLE I (Continued)

Note 1: Each column, for the condition shown, indicates a combination of requirements that can be obtained, as qualified below.

Note 2: "A" indicates a minimum requirement for Merchant Quality Hot Rolled Carbon Steel Bars.

Note 3: "N" indicates an additional requirement which must be negotiated.

Note 4: "S" indicates an additional requirement which may be specified.

Note 5: "X1" indicates one group of optional available requirements. Within that group only one option may be specified. Similarly, in group "X3", only one option may be specified.

5.1.2 Type of Steel.- The type of steel applied on Merchant Quality as determined by deoxidation practice is left to the producer's discretion. Bars of Merchant Quality shall not be specified to any silicon content, grain size, or other restriction which would influence the type of steel.

5.1.3 Description and Component Requirements.- Merchant Quality bars shall be free from visible pipe; however, they may contain pronounced chemical segregation and for this reason check analysis is not appropriate except when misapplication is clearly indicated. Internal porosity, surface seams and other surface irregularities are to be expected and are not cause for rejection. Consequently Merchant Quality is not suitable for applications such as forging, heat treating or other operations where internal defects or surface imperfections are objectionable. Resulphurized or thermally treated steel bars are not available in Merchant Quality.

Merchant Quality hot rolled bars shall be furnished as-rolled (see 3.2) or, when specified, with scale removed (see A29) or with scale removed and coated (see A30) for temporary corrosion protection. When ordered to composition, a report of ladle analysis shall be furnished for each heat of steel. When ordered to mechanical properties, a report of the mechanical test results and ladle analysis (each heat) for phosphorus and sulphur shall be furnished. Dimensional tolerances for size, length, and straightness shall be as listed in the tolerance tables (see 5.4). The bars shall be subject to visual examination for pipe.

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5.1.4 Merchant Quality Specified to Composition.

A-1 COMPOSITION

Bars specified to composition, ladle analysis only, shall be as listed in Tables II, III, and IV for the grade or ranges specified. Description and component requirements applicable to Merchant Quality are as stated in 5.1.3.

TABLE II

GRADES OF MERCHANT QUALITY

Grade	C	Mn	P	S Max
M-1008	.10 Max	.25/.60	.04	.05
M-1010	.07/.14	.25/.60	.04	.05
M-1012	.09/.16	.25/.60	.04	.05
M-1015	.12/.19	.25/.60	.04	.05
M-1017	.14/.21	.25/.60	.04	.05
M-1020	.17/.24	.25/.60	.04	.05
M-1023	.19/.27	.25/.60	.04	.05
M-1025	.20/.30	.25/.60	.04	.05
M-1031	.26/.36	.25/.60	.04	.05
M-1044	.40/.50	.25/.60	.04	.05
MB-1006	.10 Max	.45 Max	.07/.12	.06
MB-1010	.13 Max	.25/.60	.07/.12	.06

Note: MB denotes acid Bessemer Steel.

TABLE III[1/]

STANDARD AISI Steels[2/] for MERCHANT QUALITY BARS

(Where closer ranges for C and Mn are required than shown in Table II)

GRADES

1008	1015	1023	1044
1010	1017	1025	B-1006
1012	1020	1031	B-1010

[1/] Copper when specified shall be 0.20% min.

[2/] See paragraph 4.5

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TABLE IV

Ranges for Other Than Standard Grades for Merchant Quality

For Max. Carbon Limit of	Carbon Range	Manganese Range	P. Max.	S Max.
.37 to .50 incl.	.10%	.25 to .60	.04	.05
or when closer ranges are required:				
.35 to .50 incl.	.08%	.30 to .60	.04	.05

5.1.5 Merchant Quality Specified to Mechanical Properties.

A-2 TENSILE PROPERTIES

Bars specified to tensile properties shall be as listed in Table V for the class specified. Ladle analysis limits for phosphorus and sulphur shall be as listed in Table VI. Description and component requirements applicable to Merchant Quality are as stated in 5.1.3.

TABLE V

Mechanical Property Requirements

Class 1/	Tensile Strength PSI Range	Yield Point min.PSI	Elongation, min. % 8" gage 2/ 3/	2" gage 4/
45	45000-55000	22500	27	33
50	50000-60000	25000	25	30
55	55000-65000	27500	23	26
60	60000-72000	30000	21	22
65	65000-77000	32500	17	20
70	70000-85000	35000	14	18
75	75000-90000	37500	14	18
80	80000-95000	40000	13	17

ASTM A 306

1/ These tensile requirements do not apply to bars (except flats) less than 1/2" in thickness or diameter or to bar size shapes less than 1 sq. in. in cross section; however, chemistry shall be applied that is consistent with the properties desired.

2/ For material under 5/16" in thickness or diameter deduct 2.0% for each 1/32" under 5/16".

3/ For material over 3/4" in thickness or diameter deduct 0.25% for each 1/32" over 3/4", but not to exceed a total deduction of 3.0%.

4/ For material over 2" in thickness or diameter deduct 1.0% for each inch (or fraction thereof) over 2", but not to exceed a total deduction of 3%.

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TABLE VI
Phosphorus and Sulphur Limits when
Mechanical Property Requirements are Specified. 1/

	Ladle Limits of Range, percent	
	Phos.	Sul.
Open Hearth	.040 max.	.050 max.
Acid Bessemer	.07/.12	.060 max.

1/ A range in carbon shall not be specified with Mechanical Property Requirements.

A-3 TENSILE & BEND PROPERTIES

Bars specified to tensile and bend properties shall be as listed in Tables V and VII. Ladle analysis limits for phosphorus and sulphur shall be as listed in Table VI. Description and component requirements applicable to Merchant Quality are as stated in 5.1.3.

TABLE VII
Bend Test Requirements

Ratio of Bend Diameter to Thickness of Specimen 1/ 3/
For Thickness or Diameter of Bar, Inches

Class	3/4 & under	Over 3/4 to 1 incl.	Over 1 to 1-1/2 incl.	Over 1-1/2 2/ to 2 incl.	Over 2 2/ to 3 incl.	Over 3 2/ to 5	Over 5 2/
45	Flat	Flat	1/2	1	1	2	3
50	Flat	1/2	1	1-1/2	2-1/2	3	3-1/2
55	1/2	1	1-1/2	2	2-1/2	3	3-1/2
60	1/2	1	1-1/2	2-1/2	3	3-1/2	4
65	1	1-1/2	2	3	3-1/2	4	5
70	1-1/2	2	2-1/2	3	3-1/2	4	5
75	2	2	3	3-1/2	4	4-1/2	6
80	2	2-1/2	3	3-1/2	4	4-1/2	6

1/ Bend test specimens shall be bent 180° at room temperature.
2/ When machined specimens are used on sizes over 1-1/2" in thickness or diameter, the ratio of bend diameter to thickness of the bend test specimen shall be increased by 50%. (Machined specimens are described in Method 231 of Fed. Test Method Std. No. 151.)
3/ Mechanical Property Requirements for Merchant Quality are applicable only to sizes shown in 5.3.2.1.

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5.2 Special Quality.

5.2.1 Procurement guide.- Special Quality bars may be specified to Chemical Composition (ladle analysis) or to Mechanical Properties. As limited by the descriptions of the individual quality or property, Special Quality bars may also be specified

(1) with one additional requirement (grain size or thermal treatment),

(2) with one Restrictive Requirement (special surface, special soundness or special discard),

(3) with one or more Additional Requirements and Restrictive Requirements, if technologically applicable and if agreed upon with the producer,

(4) with Restricted Requirements (chemical or mechanical, size and length) if agreed upon with the producer, and

(5) for Identified Applications.

A guide to specifying requirements in Special Quality hot rolled bars is presented in Table VIII. This guide has been set up in the same manner as the one for Merchant Quality steel bars in table I and the explanation would be applicable for this table.

TABLE VIII

GUIDE TO SPECIFYING REQUIREMENTS IN SPECIAL QUALITY HOT ROLLED CARBON STEEL BARS IN THE FORM OF ROUNDS, SQUARE CORNERED SQUARES, ROUND CORNERED SQUARES, HEXAGONS, FLATS, BAR SIZE SHAPES, OVALS, HALF OVALS AND HALF ROUNDS

	Requirement	Condition					
		As Hot Rolled		Annealed,* Normalized, Stress Relieved		Quenched and Tempered	
		Chem. Comp.	Mech. Prop.	Chem. Comp.	Mech. Prop.	Chem. Comp.	Mech. Prop.
Ladle Chemical Composition	Standard Steels	X1		X1		X1	
	Other Than Standard Steels	X1		X1		X1	
	Restricted Composition Range	X1N		X1N		X1N	
	Si .10 Max.	X2	X2				
	Si .10/.20 or .15/.30	X2	X2	X2	X2	X2	X2
	Cu .20 Min.	S	S	S	S	S	S
	Pb .15/.35	S	S	S	S	S	S
	Four Elements - C, Mn, P, S	A		A		A	N
	Two Elements - P, S		A		A		A
	Incidental Elements, Controlled Max.	N	N	N	N	N	N
	Incidental Elements, Reported	S	S	S	S	S	S

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TABLE VIII (Continued)

	Requirement	Condition					
		As Hot Rolled		Annealed* Normalized Stress Relieved		Quenched and Tempered	
		Chem. Comp.	Mech. Prop.	Chem. Comp.	Mech. Prop.	Chem. Comp.	Mech. Prop.
Mechanical Properties	Tensile Test, Standard		X3		X3		X3
	Tensile Test, Restricted		X3N		X3N		X3N
	Tensile and Cold Bend Tests		X3		X3		X3
	Hardness Test				X3		X3
	Tensile Strength-Closer than ASTM Standard Ranges		X3N		X3N		X3N
	Tensile Strength, Min., Max., or Range		X3				X3
	Tensile Strength, Min. or Max.		X3		X3		X3
	Hardness, Max.				X3N		X3N
	Hardness, Min., Max., or Range .3 MM or Larger Brinell Impression Diameter						X3N
Type of Steel and Grain Size	Non-killed Steel	S	S				
	Killed Steel	S	S	S	S	S	S
	Grain Size (Austenitic), Coarse Grain 1/5 or Fine Grain 5/8 Killed Steel	S	S	S	S	S	S
Size	Standard	A	A	A	A	A	A
	Nonstandard	N	N	N	N	N	N
Cutting Length	Standard Mill Cutting	A	A	A	A	A	A
	Special Re-cutting of Bar Ends	S	S	S	S	S	S
	Random Lengths	A	A	A	A	A	A
	Specified Length - Max. Shorts 10%	A	A	A	A	A	A
	Multiple Lengths	N	N	N	N	N	N
	Specified Length - No Shorts	S	S	S	S	S	S
Tolerances	Straightness Tolerances, Standard	A	A	A	A	A	A
	Straightness Tolerances, Special	S	S	S	S	S	S
	Dimensional Tolerances, Standard	A	A	A	A	A	A
	Dimensional Tolerances, Restricted	N	N	N	N	N	N
	Length Tolerances, Standard	A	A	A	A	A	A
	Length Tolerances, Restricted	S	S	S	S	S	S
Processing	Pickled - Coated or Uncoated	S	S	S	S	S	S
	Special Preparation of One End To Special Straightness Tolerances	S	S	S	S	S	S
		S	S	S	S	S	S
Inspection	Special Inspection Practice Involving Re-handling	S	S	S	S	S	S
Report of Tests	Ladle Analysis	A	A	A	A	A	A
	Tensile Test	-	A	-	A	-	A
	Bend Test	-	A	-	A	-	A
	Hardness Test	-	-	-	A	-	A
	Specified Tests Requiring Extensometer Use	-	-	-	S	-	S
Miscellaneous	Check Analysis - Drillings For	S	S	S	S	S	S

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TABLE VIII (Continued

*When annealed condition is required, specify ordinary annealing, spheroidize annealing or lamellar pearlitic annealing.

Note 1: Each column for the condition shown, indicates a combination of requirements that can be obtained, as qualified below.

Note 2: "A" indicates a minimum requirement for Special Quality Hot Rolled Carbon Steel Bars.

Note 3: "N" indicates an additional requirement which must be negotiated.

Note 4: "S" indicates an additional requirement which may be specified.

Note 5: "X1" indicates one group of optional available requirements. Within that group only one option may be specified. Similarly, in group X2 and group X3, only one option in each group may be specified.

Note 6: "X1N" indicates one of a group of optional available requirements which must be negotiated.

5.2.2 Type of Steel.- Rimmed, capped, semikilled and killed steels may be specified to Special Quality, dependent upon chemical composition, producer's manufacturing facilities, purchaser's methods of fabrication and end use. Killed steels include steels specified to: (a) silicon minimum (b) a range of silicon, or (c) a silicon maximum over 0.10 percent. Killed steels may be specified to coarse or fine austenite grain size.

5.2.3 Description and Component Requirements.- Special Quality bars shall be free from pipe as determined by visual examination of billet and bar ends. Although the blooms or billets are inspected and prepared, as necessary, by scarfing, chipping, grinding or other means, to minimize injurious surface imperfections, Special Quality bars are likely to contain surface imperfections the frequency and severity of which are influenced by the type of steel, chemical composition and bar size. When relative freedom from surface imperfections is of prime importance, Special Surface bars, as described herein, should be specified unless the fabricating procedure involves an adequate removal of surface metal as recommended in 5.5 on Machining Allowances. Lower carbon or lower manganese grades, especially in killed steels or resulphurized steels, are most susceptible to surface imperfections. In general, the severity of surface imperfections tends to increase with bar size.

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Special Quality Bars shall be furnished as-rolled (see 3.2) or when specified, with scale removed (see A29), or with scale removed and coated (see A30) for temporary corrosion protection. When ordered to composition, a report of ladle analysis shall be furnished for each heat of steel. When ordered to mechanical properties, a report of the mechanical test results and ladle analysis (each heat) for phosphorus and sulphur shall be furnished. Dimensional tolerances for size, length and straightness shall be as listed in the tolerance tables (5.4). The bars shall be subject to check analysis tolerances (4.5.2) by the purchaser for applicable elements and to visual examination for pipe and detrimental surface imperfections.

5.2.4 Special Quality Specified to Chemical Composition.

A-4 COMPOSITION

Bars specified to chemical composition shall be in accordance with 4.5. Copper when specified shall be 0.20% minimum ladle analysis. Description and component requirements applicable to Special Quality are as stated in 5.2.3.

5.2.5 Special Quality Specified to Mechanical Properties.

A-5 TENSILE PROPERTIES

Bars specified to tensile properties shall be as listed in Table IX for the class specified. Ladle analysis limits for phosphorus and sulphur shall be as listed in Table X. Copper when specified shall be 0.20% minimum ladle analysis. Description and component requirements applicable to Special Quality are as stated in 5.2.3.

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TABLE IX 5/
Mechanical Property Requirements

Class 1/	Tensile Strength PSI Range	Yield Point Minimum PSI	Elongation, Min. % 8" gage 2/ 3/	2" gage 4/
45	45000-55000	22500	27	33
50	50000-60000	25000	25	30
55	55000-65000	27500	23	26
60	60000-72000	30000	21	22
65	65000-77000	32500	17	20
70	70000-85000	35000	14	18
75	75000-90000	37500	14	18
80	80000-95000	40000	13	17

1/ These tensile requirements do not apply to bars (except flats) less than 1/2" in thickness or diameter or to bar size shapes less than 1 sq. in. in cross section; however, chemistry shall be applied that is consistent with the mechanical properties desired.

2/ For material under 5/16" in thickness or diameter deduct 2.0% for each 1/32" under 5/16".

3/ For material over 3/4" in thickness or diameter deduct 0.25% for each 1/32" over 3/4", but not to exceed a total deduction of 3.0%.

4/ For material over 2" in thickness or diameter deduct 1.0% for each inch (or fraction thereof) over 2", but not to exceed a total deduction of 3%.

5/ Tensile strength over 80,000 PSI min. and 95,000 PSI max. may be negotiated. Restricted tensile range, yield-tensile ratio or ductility, when required as other than given in this table, shall be negotiated.

TABLE X
Phosphorus and Sulphur Limits when Mechanical Property Requirements are Specified 1/

	Ladle Limits of Range, percent Phos.	Sul.
Open Hearth	.040 max.	.050 max.
Acid Bessemer	.07/.12	.060 max.

1/ A range in carbon shall not be specified with Mechanical Property Requirements.

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A-6 TENSILE AND BEND PROPERTIES

Bars specified to tensile and bend properties shall be as listed in Tables IX and XI (see A-5). Ladle analysis limits for phosphorus and sulphur shall be as listed in Table X. Copper when specified shall be 0.20% minimum. Description and component requirements applicable to Special Quality are as stated in 5.2.3.

TABLE XI
Bend Test Requirements

Ratio of Bend Diameter to Thickness of Specimen 1/
For Thickness or Diameter of Bar, Inches

Class	3/4 & under	Over 3/4 to 1 incl.	Over 1 to 1-1/2 incl.	Over 1-1/2 2/ to 2 incl.	Over 2 2/ to 3 incl.	Over 3 2/ to 5	Over 5 2/
45	Flat	Flat	1/2	1	1	2	3
50	Flat	1/2	1	1-1/2	2-1/2	3	3-1/2
55	1/2	1	1-1/2	2	2-1/2	3	3-1/2
60	1/2	1	1-1/2	2-1/2	3	3-1/2	4
65	1	1-1/2	2	3	3-1/2	4	5
70	1-1/2	2	2-1/2	3	3-1/2	4	5
75	2	2	3	3-1/2	4	4-1/2	6
80	2	2-1/2	3	3-1/2	4	4-1/2	6

1/ Bend test specimens shall be bent 180° at room temperature.
2/ When machined specimens are used on sizes over 1-1/2" in thickness or diameter, the ratio of bend diameter to thickness of the bend test specimen shall be increased by 50%. (Machined specimens are described in Method 231 of Fed. Test Method Std. No. 151.)

5.2.6 Additional Grain Size Requirement.—Austenite Grain Size Requirements are commonly specified for applications involving carburizing or heat treatment.

A-7 AUSTENITE GRAIN SIZE

Special Quality bars specified to this requirement shall be produced to killed steel practice and shall be in accordance with the following Austenite Grain Sizes, as specified:

Coarse grain (size 1 to 5 incl.)

Fine grain (size 5 to 8 incl.)

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Variations in steel making practice or in preparation of specimens can be such that all grains do not meet a particular grain size. The grain structure is considered to be within limits if 70% of the area examined is within the specified limits.

If desirable, a check for grain size on finished bars shall be made by the purchaser.

5.2.7 Additional Thermal Treatment Requirements.- When specified, hot rolled carbon steel bars of Special Quality may be specified to thermal treatments as described herein.

Due to warpage during treatment, thermally treated bars are not normally within the standard straightness tolerance. Therefore, when necessary, such bars may be specified to Special Straightness Tolerance as defined in 5.2.11. The straightening operation may slightly harden the bars and introduce stresses which may cause distortion in subsequent machining or processing operations. If this distortion cannot be tolerated, it is necessary to specify stress relief after straightening.

A-8 ORDINARY ANNEALED

The bars shall be uniformly heated to and held at a suitable temperature above the transformation range followed by retarded cooling.

An appropriate maximum hardness value may be specified subject to agreement. It is not practicable to maintain a range of hardness.

A-9 SPHEROIDIZE ANNEALED[1/]

The bars shall be subjected to prolonged heating at the appropriate temperature followed by slow or cycle cooling, or other appropriate thermal treatment, so as to produce a globular or spheroidal condition of the carbide.

1/ This treatment improves machinability of high carbon steels; it may also be desirable for cold forming, cold drawing or its effect on subsequent heat treatments.

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An appropriate maximum hardness may be specified subject to agreement. It is not practicable to maintain a range in hardness.

A-10 LAMELLAR PEARLITIC ANNEALED[1/]

The bars shall be subjected to an appropriate heating and cooling treatment so as to produce a lamellar pearlitic structure.

An appropriate maximum hardness may be specified subject to agreement. It is not practicable to maintain a range in hardness.

Subject to agreement, the bars may be subject to micro-examination by the purchaser for lamellar pearlitic structure.

A-11 NORMALIZED

The bars shall be uniformly heated to a suitable temperature above the transformation range and then cooled in still air to room temperature. An appropriate maximum hardness value may be specified subject to agreement. It is not practicable to maintain a range of hardness.

A-12 STRESS RELIEVED

The bars shall be heated to a suitable temperature below the transformation range, held for a time interval sufficient to equalize the temperature throughout the piece and result in reduced residual stresses in the bars after cooling. An appropriate maximum hardness value may be specified subject to agreement. It is not practicable to maintain a range of hardness.

A-13 QUENCHED & TEMPERED

The bars shall be uniformly (1) heated to a suitable temperature above the transformation range to approach a uniform solid solution (Austenite) (2) hardened by quenching in water, oil, or other suitable medium, and (3) tempered by reheating to a temperature below the transformation range to secure the desired properties. The temperature selected for tempering shall not be less than 800°F. Quenched and tempered bars may be specified to a hardness range or to mechanical properties. The

[1/] This treatment improves machinability of medium and low carbon steels, except the very low carbon grades.

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hardness range specified shall be not less than 0.3 millimeter Brinell impression diameter at an identified test location. Mechanical property requirements shall be specified in accordance with Table XII. Other mechanical properties may be specified subject to agreement.

TABLE XII

Tensile Requirements

Diameter or Distance Between Parallel Faces, in.	Yield Point min, psi	Tensile Strength min, psi	Elongation in 2 in., min. percent	Reduction of Area, min. percent
1 and under	75 000	110 000	18	45
Over 1 to 2-1/2 incl.	70 000	105 000	18	45
Over 2-1/2 to 4, incl.	65 000	95 000	18	45
Over 4 to 6, incl.	60 000	90 000	18	40
Over 6 to 9-1/2, incl.	50 000	85 000	18	35

ASTM A 321

5.2.8 Restrictive requirements.- Certain applications or fabricating processes may require one or more of the Restrictive Requirements described herein for Special Quality hot rolled carbon steel bars. It is desirable to specify not more than one of the restrictive requirements referring to quality. However, more than one quality restriction may be specified by arrangement with the supplier, if the cumulative restrictions are not conflicting or technologically impractical. Other necessary restrictive requirements may be specified upon agreement.

A-14 SPECIAL SURFACE

When specified Special Quality bars to Special Surface Requirement shall be relatively free from surface imperfections and the surface in the as-rolled condition, or after further processing, may be subjected, by the purchaser, to surface inspection procedures more critical than visual examination.

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A-15 SPECIAL SOUNDNESS

When specified Special Quality bars to Special Soundness Requirement shall have a greater degree of soundness than determinable by visual examinations of billet or bar ends. The Macro-etch test per Fed. Test Method Std. No. 151 or the facture test per ASTM A273-58T shall be used to determine the degree of soundness.

A-16 SPECIAL HEAT TREATING REQUIREMENTS

This is a term generally used when the mechanical properties desired after purchaser's heat treatment are specified to the producer as a requirement. All phases of heat treatment procedure and the mechanical property requirements are subject to agreement.

5.2.9 Restricted Requirements.- Usage has developed the need by designers for certain restrictions which have resulted in economical production of parts when a product is available with these special properties. Such a product is available in Special Quality or Merchant Quality, as follows:

A-17 RESTRICTED MECHANICAL PROPERTIES

(Applicable to Special Quality)

Restricted requirement for tensile range, yield-tensile ratio or ductility is sometimes necessary. Such requirement shall be negotiated.

A-18 RESTRICTED CHEMICAL COMPOSITION

(Applicable to Special Quality)

Where closer than standard ranges and limits of chemical elements are necessary, they may be specified by agreement.

A-19 RESTRICTED SIZE TOLERANCE

(Applicable to Special Quality)

When necessary, closer than standard cross sectional tolerances may be specified by agreement. (For standard tolerances, see 5.4.)

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A-20 RESTRICTED LENGTH TOLERANCE

(Applicable to all Qualities)

When it is necessary to specify a product to a length with closer than standard cutting, the tolerances as listed in reference la12 may be specified. Such tolerances can be specified only if special straightness is also specified.

5.2.10 Identified Quality Applications.- Carbon steel bars may be specified to certain well established quality designations intended for Specific applications, as described herein. The individual qualities necessary for each identified application have developed by usage.

A-21 AXLE SHAFT QUALITY

Bars of this quality may be specified for use in the manufacture of power driven axle shafts of the automotive and truck type. The bars shall be produced to rigorous surface and internal quality requirements. Chemical Composition shall be in accordance with 4.5. Typical Standard Steel Compositions are:

AISI-1035, 1037, 1038, 1040, 1041, 1045, 1046

A-22 COLD HEADING & COLD FORGING QUALITY

Bars of this Quality are subject to cold plastic deformation by upsetting, heading or forging. When the maximum specified carbon is 0.30% or over, or manganese is 1.20% or over, the bars should be specified ordinary annealed, or spheroidized annealed depending on the severity of the operation. The bars shall have internal soundness and no injurious surface imperfections. When the specific end use (part) is made known to the producer, the bars shall be subject to the satisfactory fabrication of the part by the purchaser using reasonable care and fabricating technique. Typical standard steel compositions are:

1008, 1010, 1016, 1018, 1020, 1022, 1035, 1038, 1041, 1045

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A-23 COLD EXTRUSION QUALITY

Bars for Cold Extrusion are subject to severe cold plastic deformation, by forward extrusion, backward extrusion or upsetting into a variety of hollow or solid shapes. Die design is a most important factor and should be such that the part is under compression during forming, otherwise tension stresses will impose a severe burden on the steel, occasionally to the extent of causing fracture in sound metal.

Some cold extruding processes are relatively new and in a rapid stage of development. Until recently, many parts were produced in a series of operations on presses, with intermediate anneals that were necessitated by the chemical composition of the steel, total amount of cold deformation, designs of the part or usage of the final product. Recent developments include forming in a series of operations in a multiple stage header, with no intermediate anneal.

Steel for these applications must be produced with closely controlled steel making practices and rigid testing and inspection for internal soundness and surface imperfections.

When annealing is necessary on account of, heavy cold work in the first stage, impossibility of intermediate annealing or when the steel does not have satisfactory cold shaping characteristics in the as-rolled condition, the purchaser should provide for, or specify a suitable type of annealing to the steel producer.

If hot shaping precedes the cold extrusion operation, the steel quality requirements can be similar to those for processes involving only cold extrusions.

Typical Standard Steel Compositions are:

1008, 1010, 1011, 1012, 1018, 1019, 1024, 1027, 1035

A-24 RESTRICTIVE COLD WORKING QUALITY

Some finished products manufactured by cold heading, cold forging, or cold extrusion have shapes and requirements on the finished article that necessitate the use of a more restrictive quality of steel. Such products include more difficult forming operations, more difficult continuous operations where intermediate annealing is impossible, parts for which the steel cannot be kept entirely under compression and have some tension stresses present, or products with specified final properties such as restrictive surface or mechanical property requirements.

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When work hardening must be minimized and when severe stress develops in the more difficult operations, a fully killed fine grained steel is usually required produced with closely controlled steel making practice and rigid testing and inspection for internal soundness and surface imperfections. When annealing of the bars is necessary the purchaser should specify tye type of annealing desired.

A-25 COLD PUNCHED NUT QUALITY

Hot rolled round bars of this quality are cold rolled or otherwise cold reduced by the purchaser to flat or rectangular cross sections from which nuts are produced by cold punching process. The bars shall be produced with no injurious surface defects and shall be satisfactory for the intended application, using reasonable care and processing techniques.

Typical standard steel compositions are:

1108, 1109, 1110

A-26 SCRAPLESS NUT QUALITY

Round bars of this quality are used to produce nuts by cold upsetting, piercing and forming, with consequent expansion in diameter. If the type of steel or chemical composition specified does not have satisfactory cold forming characteristics in the as-rolled condition, normalizing, annealing, or spheroidizing annealing may be necessary. The bars shall be produced to rigorous surface and internal soundness requirements and shall be satisfactory for the intended application, using reasonable care and processing techniques.

Typical standard steel compositions are:

1008, 1009, 1010, 1018, 1020, 1023, 1030, 1038, 1109

5.2.11 Special Processing Requirements.

A-27 SPECIAL STRAIGHTNESS

When regular straightness tolerances are not adequate, bars may be specified to special straightness tolerance as defined in the tolerance table (5.4).

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A-28 SPECIAL RECUTTING OF BAR ENDS

When the end distortion or burrs, normally encountered in regular mill cutting methods, are not satisfactory to the purchaser, one or both ends of the bar can be recut by abrasive, machine or torch cutting, or other means dependent upon facilities available in the producer's plant.

A-29 SCALE REMOVAL

When specified, hot rolled bars and thermally treated bars are descaled by pickling, grit blasting, or other means, dependent upon the facilities at the producer's plant.

A-30 SURFACE COATING

In general, descaled bars should be protected from rust with a surface coating in order to be suitable for shipment and short-time storage. In some cases bars which have not been descaled are required to be protected. Care should be used in selecting a specified coating that will not adversely affect later processing. Subject to agreement, these bars may be furnished with a special coating for further processing. When specified, bar surfaces are coated with oil, lime, or special materials which are recognized as mild rust inhibitors. Because such surface coatings are intended to be only temporary protection against rust, extended storage or storage in moist or contaminated atmospheres requires further precautionary measures against the accumulation of condensation or other forms of moisture.

5.3 Sizes and size ranges.

5.3.1 Sizes.- Hot rolled carbon steel bars shall be furnished to the nominal sizes as specified in the invitation for bids, contract or order. Tables XIII through XXX show those nominal sizes that are currently in general production and demand, and afford an adequate selection of sections for ordinary uses and for stocks. If sizes other than those listed are required, the producer should be consulted before specification.

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Table XIII

Rounds for all purposes

Nominal Sizes inclusive (inches)	Fractional Increments (inches)
1/4 to 29/32	Advancing by sixty-fourths
Over 29/32 to 2-1/16	Advancing by thirty-seconds
" 2-1/16 to 4-1/8	Advancing by sixteenths
" 4-1/8 to 6-1/4	Advancing by eights
" 6-1/4 to 10	Advancing by fourths

Table XIV

Rounds for bolts, rivets, etc.[1/]

Decimal Sizes (inches)		
0.365	0.865	1.235
0.445	0.912	1.297
0.490	0.990	1.360
0.615	1.047	1.422
0.680	1.110	1.485
0.740	1.172	

1/ Diameters listed in tables XIV and XV are sizes intermediate to those listed fractionally in Table XIII. However, many diameters listed in Table XIII are also regularly ordered by their decimal equivalents, for bolt and rivet stock or for general purposes.

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Table XV
Rounds for heat-treated studs, etc. 1/

Decimal Sizes (inches)	
0.507	1.135
0.632	1.261
0.758	1.387
0.883	1.514
1.009	

1/ See footnote 1, Table 2a.

Table XVI

Squares

Nominal Sizes inclusive (inches)	Fractional Increments (inches)
1/4 to 1-5/16	Advancing by thirty-seconds
1-5/16 to 4-1/4	Advancing by sixteenths
4-1/4 to 5-1/2	Advancing by fourths

Table XVII
Round-cornered squares 1/

Nominal Sizes, inclusive (inches)	Fractional increments (inches)	Nominal Corner-radii (inches)
3/8 to 1/2	Advancing by thirty-seconds	1/16
17/32 to 13/16	Advancing by thirty-seconds	3/32
27/32 to 1-15/32	Advancing by thirty-seconds	1/8
1-1/2 to 1-15/16	Advancing by sixteenths	1/4
2 to 2-7/16	Advancing by sixteenths	5/16
2-1/2 to 2-7/8	Advancing by sixteenths	3/8
3 to 3-3/8	Advancing by eighths	7/16
3-1/2 to 3-7/8	Advancing by eighths	1/2
4 to 4-1/4	Advancing by fourths	5/8
4-1/2 to 5-1/2	Advancing by fourths	3/4

1/ Sizes are face to face. Round-cornered squares shall be rolled to dimensions, not to weights per linear foot.

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Table XVIII

Hexagons

Nominal Sizes, inclusive (inches)	Fractional Increments (inches)
1/4 tto 2-1/16	Advancing by thirty-seconds
2-1/16 to 4-1/16	Advancing by sixteenths

Table XIX

Half-rounds

Nominal Sizes, inclusive (inches)	Fractional increments (inches)
3/8 to 1	Advancing by eighths
1 to 2	Advancing by fourths
2-1/2 and 3	

Table XX

Ovals, sharp

Nominal Sizes (inches)
3/8 by 3/16
1/2 by 1/4
5/8 by 5/16
3/4 by 5/16
7/8 by 7/16
1 by 7/16
1-1/4 by 7/16
1-1/2 by 7/16
1-3/4 by 7/16
2 by 7/16

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Table XXI

Ovals, blunt

Nominal Sizes (inches)
3/8 by 1/8
7/16 by 1/8
1/2 by 1/4
9/16 by 1/4
5/8 by 1/4

Table XXII

Half-ovals

Nominal Sizes (inches)								
3/8 by 3/32								
1/2 by	1/8	3/16						
5/8 by	1/8	3/16	1/4					
3/4 by		3/16	1/4					
7/8 by		3/16	1/4	5/16				
1 by		3/16	1/4	5/16	3/8			
1-1/8 by		3/16	1/4	5/16	3/8			
1-1/4 by		3/16	1/4	5/16	3/8			
1-3/8 by			1/4	5/16	3/8			
1-1/2 by			1/4	5/16	3/8	7/16		
1-3/4 by				5/16	3/8	7/16		
2 by					3/8	7/16	1/2	
2-1/4 by						7/16	1/2	5/8
2-1/2 by							1/2	3/4
3 by								7/8
3-1/2 by								

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Table XXIII
Fillet angles, equal leg

Nominal Sizes (inches)	Thicknesses (inches)				
1/2 by 1/2	1/8				
5/8 by 5/8	1/8				
3/4 by 3/4	1/8				
7/8 by 7/8	1/8				
1 by 1	7/64	1/8	3/16	1/4	
1-1/8 by 1-1/8		1/8	3/16		
1-1/4 by 1-1/4	7/64	1/8	3/16	1/4	
1-1/2 by 1-1/2		1/8	3/16	1/4	
1-3/4 by 1-3/4		1/8	3/16	1/4	
2 by 2		1/8	3/16	1/4	5/16	3/8	
2-1/2 by 2-1/2			3/16	1/4	5/16	3/8	1/2

Table XXIV
Fillet angles, unequal leg

Nominal Sizes (inches)	Thicknesses (inches)				
1/2 by 5/16	3/32				
1 by 5/8		1/8			
1 by 3/4		1/8			
1-1/4 by 1		1/8	3/16		
1-9/32 by 29/32	5/32				
1-3/8 by 7/8		1/8	3/16		
1-1/2 by 3/4		1/8			
1-1/2 by 1		1/8			
1-1/2 by 1-1/4		1/8	3/16		
1-3/4 by 1		1/8	3/16		
1-3/4 by 1-1/4		1/8	3/16	1/4	
2 by 1-1/4		1/8	3/16	1/4	
2 by 1-1/2		1/8	3/16	1/4	
2-1/4 by 1-1/2			3/16	1/4	
2-1/2 by 1-1/2			3/16	1/4	5/16	
2-1/2 by 2			3/16	1/4	5/16	3/8

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Table XXV
Square root angles, equal leg

Nominal sizes (inches)	Thickness (inches)		
3/4 by 3/4	1/8		
7/8 by 7/8	1/8		
1 by 1	1/8		
1-1/4 by 1-1/4	1/8	3/16	
1-1/2 by 1-1/2	1/8	3/16	1/4
1-3/4 by 1-3/4		3/16	1/4
2 by 2			1/4

Table XXVI
Square root angles, unequal leg

Nominal Sizes (inches)	Thicknesses		
2 by 1	1/8	3/16	
2 by 1-1/4			1/4

Table XXVII
Channels, square back

Nominal sizes (inches)					
Width	Flanges	Web			
1/2	1/4	1/8			
3/4	5/16	1/8			
3/4	3/8	1/8			
3/4	3/8		0.072		
7/8	3/8	1/8			
7/8	7/16	1/8			
1	3/8	1/8			
1	1/2	1/8			
1-1/8	3/8	1/8			
1-1/8	9/16		3/16		
1-1/4	1/2	1/8			
1-1/2	1/2	1/8			
1-1/2	9/16		3/16		
1-1/2	11/16				1/4
1-1/2	3/4	1/8	3/16		
1-3/4	1/2		3/16		
2	1/2	1/8			
2	9/16		3/16		
2	5/8				1/4
2	1	1/8	3/16		
2-1/2	15/32	1/8			
2-1/2	5/8		3/16		

(NOTE: For the reason that some of the above items are available in more than one pattern, producers' lists should be consulted.)

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Table XXVIII
Channels, box

Nominal size (inches)		
Width	Flanges	Web
1-1/2	1-1/2	3/16

Table XXIX
Tees, equal

Nominal sizes (inches) 1/				
Flange width	Stem Depth	Thicknesses		
3/4	3/4	1/8		
7/8	7/8	1/8		
1	1	1/8	3/16	
1-1/8	1-1/8	1/8		
1-1/4	1-1/4	1/8	3/16	1/4
1-1/2	1-1/2		3/16	1/4
1-3/4	1-3/4		3/16	1/4
2	2		1/4	5/16
2-1/4	2-1/4		1/4	
2-1/2	2-1/2		1/4	5/16 3/8

1/ Dimensions of equal tees, above, and of unequal tees (Table 18a) are over-all measurements of flange and stem, and thickness of both flange and stem at outer ends, flange and stem to have the same thickness.
(NOTE: Profiles as made by different mills may differ in detail and producers' lists should be consulted.)

Table XXX
Tees, unequal

Nominal sizes (inches) 1/			
Flange width	Stem depth	Thicknesses	
1-1/2	1-1/4	1/8	
2	1-1/2	3/16	1/4
2-1/2	1-23/32	3/16	1/4

1/ See footnote 1/, Table 17a.

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5.3.2 Size Ranges. 1/ 2/ - Hot Rolled Bars are normally produced in straight lengths, but can be produced in coils dependent on size, section, and steel mill equipment. Production is in the following size ranges:

5.3.2.1 Merchant Quality.-

Rounds, squares and hexagons with diameters or distance across flats	-	under 3 inches
Bar size shapes with maximum dimensions	-	under 3 inches
Other bar sections with weight per foot	-	under 40.8 pounds per ft.
Flats 6 inches or under in width 13/64 inch and over in thickness	-	under 40.8 pounds per ft.
Flats over 6 to 8 inches inclusive in width, 0.2300 inch and over in thickness	-	under 40.8 pounds per ft.

5.3.2.2 Special or Higher Quality.

Rounds	3/8" to 10" incl.
Square Cornered Squares	3/8" to 5-1/2" incl.
Round Cornered Squares	3/8" to 8" incl.
Hexagons	3/8" to 4-1/16" incl.

Flats 13/64" and over in specified thickness, and specified widths - up to 6" incl.

Flats .230" and over in specified thickness, and specified widths over 6" to 8" incl.

Bar Size Shapes, including Angles, Channels, tees and Zees with greatest dimension - under 3"

Ovals, Half Ovals and Half Rounds (Consult producer for sizes available).

5.3.2.3 Special Bar Sections.- These sections require preparation of specific rolls, designed for the section specified, and are available only by arrangement with the producer.

1/ All producers do not make all sections or all sizes within the limits shown

2/ Refer to Tables of Standard Sizes

5.4 Tolerances.- Tolerances for hot rolled carbon steel bars and bar size shapes shall be as listed under the following identification symbols

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Identification Symbol
1a1

HOT ROLLED CARBON STEEL BARS

Rounds and Squares and Round-Cornered Squares

Size and Out-of-Round or Out-of-Square Tolerances

Specified Sizes, in.	Size Tolerances, inch		Out-of-Round or Out-of-Square Section, in.
	Over	Under	
To 5/16 incl.	0.005	0.005	0.008
Over 5/16 to 7/16 incl.	0.006	0.006	0.009
Over 7/16 to 5/8 incl.	0.007	0.007	0.010
Over 5/8 to 7/8 incl.	0.008	0.008	0.012
Over 7/8 to 1 incl.	0.009	0.009	0.013
Over 1 to 1 1/8 incl.	0.010	0.010	0.015
Over 1 1/8 to 1 1/4 incl.	0.011	0.011	0.016
Over 1 1/4 to 1 3/8 incl.	0.012	0.012	0.018
Over 1 3/8 to 1 1/2 incl.	0.014	0.014	0.021
Over 1 1/2 to 2 incl.	1/64	1/64	0.023
Over 2 to 2 1/2 incl.	1/32	0	0.023
Over 2 1/2 to 3 1/2 incl.	3/64	0	0.035
Over 3 1/2 to 4 1/2 incl.	1/16	0	0.046
Over 4 1/2 to 5 1/2 incl.	5/64	0	0.058
Over 5 1/2 to 6 1/2 incl.	1/8	0	0.070
Over 6 1/2 to 8 1/4 incl.	5/32	0	0.085
Over 8 1/4 to 9 1/2 incl.	3/16	0	0.100
Over 9 1/2 to 10 incl.	1/4	0	0.120

NOTE: Out-of-round is the difference between the maximum and minimum diameters of the bar, measured at the same cross section. Out-of-square section is the difference in the two dimensions at the same cross section of a square bar between opposite faces.

Identification Symbol
1a2

HOT ROLLED CARBON STEEL BARS

Hexagons

Size and Out-of-Hexagon Tolerances

Specified Sizes between Opposite Sides, in.	Size Tolerances, inch		Out of Hexagon Section, in.
	Over	Under	
To 1/2 incl.	0.007	0.007	0.011
Over 1/2 to 1 incl.	0.010	0.010	0.015
Over 1 to 1 1/2 incl.	0.021	0.013	0.025
Over 1 1/2 to 2 incl.	1/32	1/64	1/32
Over 2 to 2 1/2 incl.	3/64	1/64	3/64
Over 2 1/2 to 3 1/2 incl.	1/16	1/64	1/16

NOTE: Out of hexagon section is the greatest difference between any two dimensions at the same cross section between opposite faces.

Identification Symbol.
1a3 Hot Rolled Carbon Steel Bars

Square-Edge and Round-Edge Flats
Thickness and Width Tolerances

Specified Widths, in.	Thickness Tolerances, for Thicknesses Given. Over and Under, in..							Width Tolerances, in.	
	.203 to .230 excl.	.230 to 1/4 excl.	1/4 to 1/2 incl.	Over 1/2 to 1 incl.	Over 1 to 2 incl.	Over 2 to 3 incl.	Over 3	Over	Under
To 1, incl.	0.007	0.007	0.008	0.010	-	-	-	1/64	1/64
Over 1 to 2 incl.	0.007	0.007	0.012	0.015	1/32	-	-	1/32	1/32
" 2 to 4 "	0.008	0.008	0.015	0.020	1/32	3/64	3/64	1/16	1/32
" 4 to 6 "	0.009	0.009	0.015	0.020	1/32	1/16	1/16	3/32	1/16
" 6 to 8 "	*	0.015	0.016	0.025	1/32	1/16	**	*	*‡

** For flats over 6" to 8", incl. in width, and over 3" in thickness, consult producers for thickness and width tolerances.

* Flats over 6" to 8" incl. in width are not available as hot rolled carbon steel bars in thickness under 0.230.

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Identification Symbol

1a4

HOT ROLLED CARBON STEEL BARS

Bar Size Angles

Thickness, Length and Out-of-Square Tolerances

Specified Length of Leg, in.	Thickness Tolerances, for Thicknesses Given, Over and Under, in.			Tolerances for length of leg, Over and Under, in.
	To 3/16 Incl.	Over 3/16 to 3/8 Incl.	Over 3/8	
To 1, incl.	0.008	0.010	—	1/32
Over 1 to 2, incl.	0.010	0.010	0.012	3/64
Over 2 to 3 excl.	0.012	0.015	0.015	1/16

The longer leg of an unequal angle determines the size for tolerance. The out-of-square tolerance in either direction is 1½ degrees.

Identification Symbol

1a5

HOT ROLLED CARBON STEEL BARS

Bar Size Channels

Depth of Section, Width of Flange

Thickness of Web and Out-of-Square Tolerances

Specified Size of Channel, in.	Size Tolerances, Over and Under, in.				Out of square[1] of either flange, in. per inch of flange width
	Depth of Section	Width of Flanges	Thickness of Web for Thickness Given		
			To 3/16, incl.	Over 3/16	
To 1½, incl.	1/32	1/32	0.010	0.015	1/32
Over 1½ to 3, excl.	1/16	1/16	0.015	0.020	1/32

[1] For channels 5/8 in. and under in depth, the out-of-square tolerance is 3/64 in. per inch of depth.

Measurements for depth of section and width of flanges are overall.

Identification Symbol

1a6

HOT ROLLED CARBON STEEL BARS

Bar Size Tees

Dimensional Tolerances

Specified Size of Tee, in.	Width or Depth, in.		Thickness of Flange, in.		Thickness of Stem, in.		Stem Out of Square[1], in.
	Over	Under	Over	Under	Over	Under	
To 1¼, incl.	3/64	3/64	0.010	0.010	0.005	0.020	1/32
Over 1¼ to 2, incl.	1/16	1/16	0.012	0.012	0.010	0.020	1/16
Over 2 to 3, excl.	3/32	3/32	0.015	0.015	0.015	0.020	3/32

[1] Stem out-of-square is the variation from its true position of the center line of stem measured at the point.

The longer member of an unequal tee determines the size for tolerances.

Measurements for both width and depth are overall.

Identification Symbol

1a7

HALF-ROUNDS, OVALS, HALF-OVALS AND OTHER SPECIAL BAR SIZE SECTIONS

Due to the facilities available to the producer, tolerances for half-rounds, ovals, half-ovals and other special bar size sections vary among the producers.

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Identification Symbol
la8[1]/

MINIMUM RANDOM LEEWAYS REQUIRED

Specified length ft.	Min. Leeway
Min. length over 40-45 inch	5 ft. spread
Min. length over 30-40 inch	4 ft. spread
Min. length over 20-35 inch	3 ft. spread
Min. length over 5-20 inch	2 ft. spread

Identification Symbol
la9[1]/

Specific Lengths with Shorts

When specific lengths are required, the billet weight cannot always be controlled to cut the full product to specified length. Accordingly, for all bars and bar size shapes ordered to specific length with shorts, 10 percent may be furnished in random short lengths. The length tolerances of lall apply to the specific lengths. No length tolerances apply to the random shorts.

Identification Symbol
la10[1]/

Specific Lengths with No Shorts

When specific lengths with no shorts are ordered, the length tolerances of lall apply.

1/ Method for specifying cut length of bars.- In order to apply the acceptable product of the bloom or billet, it is necessary to specify one of two generally accepted methods. A third restricted method may also be used.

(a) Random lengths, as defined in Table la8, may be specified.

(b) Specific lengths plus allowable percent of shorts, as defined in Table la9, may be specified.

(c) Specific lengths with no shorts (restricted) as defined in Table la10 may be specified.

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Identification Symbol

1all - Length Tolerances - Mill Cutting

Specified Size of Rounds, Squares and Hexagons, inches.	Specified Size of Flats, Inches. Thickness	Width	Tolerances Over Specified Length, in. No Tolerances Under. 5-10 ft. excl.	10-20 ft. excl.	20-30 ft. excl.	30-40 ft. excl.	40-60 ft. incl.
to 1, incl.	To 1, incl.	To 3, incl.	1/2	3/4	1-1/4	1-3/4	2-1/4
Over 1 to 2 incl.	Over 1	To 3, incl.	5/8	1	1-1/2	2	2-1/2
Over 1 to 2 incl.	To 1, incl.	Over 3 to 6 incl.	5/8	1	1-1/2	2	2-1/2
Over 2 to 5 incl.	Over 1	Over 3 to 6 incl.	1	1-1/2	1-3/4	2-1/4	2-3/4
Over 5 to 10 incl.	-	-	2	2-1/2	2-3/4	3	3-1/4
-	.230 to 1 incl.	Over 6 to 8 incl.	3/4	1-1/4	1-3/4	3-1/2	4
-	Over 1 to 3 incl.	Over 6 to 8 incl.	1-1/4	1-3/4	2	3-1/2	4
Bar Size Sections			5/8	1	1-1/2	2	2-1/2

Note: For flats over 6" to 8" incl. in width and over 3" in thickness consult producer for length tolerances.

HOT SAWING

#2 to 5, incl.	1 & over	3 & over	#	1-1/2	1-3/4	2-1/4	2-3/4
Over 5 to 10, incl.	-	-	*	2-1/2	2-3/4	3	3-1/4

Smaller sizes and shorter lengths are not hot sawed.

Identification Symbol

1a12

Hot Rolled Carbon Steel Bars

Recutting After Straightening

Restricted Length Tolerances

Sizes of Rds. Sq. & Hexs.; Width of Flats; and Max. Dimension of Other Sections, in.	Tolerances for Specified Length, in. to 12 ft., incl. Over	Under	Over 12' to 25' incl. Over	Under
To 3, incl.	3/16	1/16	1/4	1/16
Over 3 to 6, incl.	1/4	1/16	3/8	1/16
Over 6 to 8, incl.	3/8	1/16	1/2	1/16
Rounds over 8"to 10" incl.	1/2	1/16	5/8	1/16

NOTE: For Flats over 6" to 8" incl. in width, and over 3" in thickness, consult producers for Length Tolerances.

Tolerances are sometimes required all over or all under the specified Length, in which case the sum of the two tolerances applies.

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Identification Symbol
1a13

HOT ROLLED CARBON STEEL BARS

Straightness Tolerances

All Bars and Bar Size Sections

¼ in. in any 5 ft. or ¼ in. x number of ft. of length divided by 5.

Because of warpage, straightness tolerances do not apply to bars if any subsequent heating operation or controlled cooling has been performed.

Identification Symbol
1a14

HOT ROLLED CARBON STEEL BARS
(Including Thermally Treated Bars)
Special Straightness Tolerances

All Bars and Bar Size Sections

⅛ in. in any 5 ft. or ⅛ in. x number of ft. of length divided by 5.

Because of warpage, straightness tolerances do not apply to bars if any subsequent heating operation has been performed after straightening.

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5.5 Allowance for machining of bars Turned on Centers.- Table XXXI covers non-resulphurized steels and contains the recommended increase in diameter for bars that are to be turned on centers and does not apply to bars that are to be centerless turned.

As stated previously, special quality bars are likely to contain some surface imperfections, as well as surface decarburization. Experience has shown that it is advisable for the purchaser to make adequate allowances for finishing by machining and to specify hot rolled sizes accordingly. These allowances take into consideration bar size, straightness, size tolerance and out of round tolerance and should not be interpreted as a seam depth tolerance. Bars are commonly straightened before machining.

It is recognized that centerless turning of straightened bars accomplishes more uniform removal of surface metal and as a result a lesser amount than the tabular value may prove adequate. On the other hand, resulphurized grades are typified by a poorer surface condition, and removal of a greater amount than the tabular value is recommended.

TABLE XXXI

Hot Rolled Carbon Steel Bars Recommended

Machining Allowances

Nominal Diameter, Range, Inches	Recommended Minimum Stock Removal from Diameter Inch
to 7/8 incl.	0.050
Over 7/8 to 1	.056
" 1 to 1-1/8	.061
" 1-1/8 to 1-1/4	.068
" 1-1/4 to 1-3/8	.074
" 1-3/8 to 1-1/2	.080
" 1-1/2 to 2	.106
" 2 to 2-1/2	.130
" 2-1/2 to 3-1/2	.180
" 3-1/2 to 4-1/2	.230
" 4-1/2 to 5-1/2	.280
" 5-1/2 to 6-1/2	.330
" 6-1/2 to 8-1/4	.418
" 8-1/4 to 9-1/2	.480
" 9-1/2 to 10	.506

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5.6 Test methods.- Testing shall be conducted by methods described in Fed. Test Method Std. No. 151, as follows

	Test Method
Chemical Composition	111, 112
Tensile	211
Bend	231
Brinell Hardness	242
Rockwell Hardness	243
Austenite Grain Size	311

5.7 Sampling for tests.- When more than one kind of test is involved in a lot, the samples of one test may also be used, if appropriate, as the samples for another test, e.g., bend test samples may be used for obtaining hardness; tensile test sample for chemical analysis, etc. When tests are required, the sampling shall be as described below.

5.7.1 Check analysis.- At least one sample for check analysis shall be taken from each lot in accordance with the procedure desbribed and limited in 3.4, Verification of Chemical Composition, and Procedures for Obtaining Check Analysis Samples of Fed. Std. No. 66.

5.7.2 Tension tests.- Two samples for tension tests shall be taken from each lot, unless the finished material from a lot is less than 30 tons when one tension test will be sufficient. If, however, material from one lot differs 3/8" or more in thickness, one tension test shall be made from both the thickest and the thinnest material rolled, regardless of the weight represented. The location of test specimens shall be as shown in table XXXII.

Table XXXII - Location of tension test specimens

Form	Size (Diameter of thickness of bar; width of flat)	Location of axis of test specimen
Bars, Flats	Not over 1-1/2 inches	Coincides with central axis of piece
Bars	Over 1-1/2 inches	One-fourth the diameter or thickness from the surface, and parallel to the direction of rolling
Flats	Over 1-1/2 inches	Test bars located at one-fourth width, and parallel to direction of rolling.

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5.7.3 Bend tests.- One sample for bend test shall be taken from each lot of less than 30 tons and two samples shall be taken from each lot of 30 or more tons. If, however, material from one lot differs 3/8" or more in thickness, one bend test shall be made from both the thickest and the thinnest material rolled, regardless of the weight represented.

5.7.4 Macro-etch tests.- The location, number of tests, area of the specimen to be examined and interpretation of tests are subject to agreement.

5.7.5 Hardness tests.- Two samples for hardness tests shall be taken from each lot.

5.7.6 Fracture tests.- The location, number of sections fractured, details of testing technique and acceptance limits are subject to agreement.

5.8 Lot descriptions.

5.8.1 For hot rolled condition.- A lot of hot rolled carbon steel bars in the as-rolled condition consists of all the bars of one heat or blow of steel, in each presentation of material for inspection.

5.8.2 For thermally treated condition.- A lot of thermally treated hot rolled carbon steel bars shall consist of all the bars of one heat or blow and one size or thickness if continuously treated. If batch treated, each batch of one heat or blow of one size or thickness, shall be considered one lot. To be considered as one lot all steel must be presented for inspection at one time.

5.8.3 For check analysis.- A lot for check analysis shall consist of each heat or blow submitted for inspection at one time.

5.9 Inspection.- The inspector representing the purchaser shall have free entry, at all times while work on the contract of the purchaser is being performed, to all parts of the manufacturer's works that concern the manufacture of the material ordered. The manufacturer shall afford the inspector, without charge, all reasonable facilities to satisfy him that the material is being furnished in accordance with this standard. All tests (except check analysis or grain size) and inspection shall be made at the place of manufacture prior to shipment, and shall be so conducted as not to interfere unnecessarily with the operation of the works.

5.9.1 Restricted inspection (Applicable to Special Quality).- Inspection requiring additional handling may be specified as an additional requirement.

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5.9.2 Reinspection.- A lot , which has been rejected for inspection, ... be resubmitted if properly reprocessed or all defective material removed.

5.9.3 Retests.- Retests will be permitted in accordance with Fed. Test Method Std. No. 151.

5.10 Rejection.- Any rejection shall be reported to the manufacturer within five working days from the receipt of samples by the purchaser.

5.10.1 Samples that represent rejected material shall be preserved for two weeks from the date rejection is reported to the manufacturer. In case of dissatisfaction with the results of the tests, the manufacturer may make claim for a rehearing within that time.

5.10.2 Material that shows injurious defects subsequent to its acceptance at the purchaser's works will be rejected, and the manufacturer shall be notified.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Asministration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.

(Copies of ASTM A273-58T may be purchased directly from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

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